

GBS BI HUB

HR Attrition Analysis

BI Analyst Case Study — Findings, Model & Recommendations

Power BI Star Schema · DAX Measures · Outlier & Trend Analysis · SQL

Executive Summary

Out of 1,470 employees, 237 left the company — an overall attrition rate of 16.1%. The strongest drivers of attrition are overtime workload, low entry-level compensation, and role-specific pressure in Sales.

Employees who leave earn roughly 30% less, have 35% shorter tenure with their current manager, and are on average 4 years younger than those who stay. This report documents the Power BI data model, DAX logic, statistical findings, and actionable recommendations.

1. Data Model — Star Schema

The Power BI model follows a Star Schema: one central Fact table connected to eight Dimension tables. The grain of the model is one row per employee. This structure keeps the fact table lean and numeric, supports fast filtering, and avoids many-to-many relationship issues.

Fact_Employee (numeric / measurable columns + foreign keys)

Age, DailyRate, DistanceFromHome, Education, EmployeeNumber (PK), EnvironmentSatisfaction, HourlyRate, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, plus the 8 foreign keys below.

Dimension Tables (8)

Dimension Table	Columns
Dim_Attrition	AttritionKey, Attrition
Dim_BusinessTravel	TravelKey, BusinessTravel
Dim_Department	DepartmentKey, Department
Dim_EducationField	EducationFieldKey, EducationField
Dim_Gender	GenderKey, Gender
Dim_JobRole	JobRoleKey, JobRole
Dim_MaritalStatus	MaritalStatusKey, MaritalStatus
Dim_OverTime	OverTimeKey, OverTime

*Rule applied throughout the model: text/categorical columns (Department, Gender, JobRole, etc.) are never stored in the Fact table — only their surrogate keys. All relationships are 1 (dimension) → * (fact), single direction.*

2. DAX Measures

15 measures were created using variables (VAR/RETURN), organized into four display folders inside the model.

Core Metrics

```
Total Employees = COUNTROWS(Fact_Employee)

Attrition Count = CALCULATE(COUNTROWS(Fact_Employee), Dim_Attrition[Attrition] = "Yes")

Attrition Rate % = DIVIDE([Attrition Count], [Total Employees], 0)
```

Income & Tenure

```
Avg Monthly Income (Leavers) =  
VAR LeaversTable = CALCULATETABLE(Fact_Employee, Dim_Attrition[Attrition] = "Yes")  
RETURN AVERAGEX(LeaversTable, Fact_Employee[MonthlyIncome])  
  
Avg Monthly Income (Stayers) =  
VAR StayersTable = CALCULATETABLE(Fact_Employee, Dim_Attrition[Attrition] = "No")  
RETURN AVERAGEX(StayersTable, Fact_Employee[MonthlyIncome])  
  
Tenure Gap (Stayers - Leavers) =  
VAR AvgStay = CALCULATE(AVERAGE(Fact_Employee[YearsAtCompany]),  
Dim_Attrition[Attrition]="No")  
VAR AvgLeft = CALCULATE(AVERAGE(Fact_Employee[YearsAtCompany]),  
Dim_Attrition[Attrition]="Yes")  
RETURN AvgStay - AvgLeft
```

OverTime Analysis

```
OverTime Attrition Rate % =  
VAR OTBase = CALCULATE([Total Employees], Dim_OverTime[OverTime]="Yes")  
VAR OTLeft = CALCULATE([Attrition Count], Dim_OverTime[OverTime]="Yes")  
RETURN DIVIDE(OTLeft, OTBase, 0)  
  
No OverTime Attrition Rate % =  
VAR NoOTBase = CALCULATE([Total Employees], Dim_OverTime[OverTime]="No")  
VAR NoOTLeft = CALCULATE([Attrition Count], Dim_OverTime[OverTime]="No")  
RETURN DIVIDE(NoOTLeft, NoOTBase, 0)
```

Risk Segments

```
Highest Risk JobRole Attrition % =  
CALCULATE([Attrition Rate %], Dim_JobRole[JobRole] = "Sales Representative")  
  
JobLevel 1 Attrition % =  
CALCULATE([Attrition Rate %], Fact_Employee[JobLevel] = 1)  
  
Zero Stock Option Attrition % =  
CALCULATE([Attrition Rate %], Fact_Employee[StockOptionLevel] = 0)  
  
High Risk Segment =  
VAR CurrentRate = [Attrition Rate %]  
RETURN IF(CurrentRate >= 0.20, "High Risk", IF(CurrentRate >= 0.15, "Medium", "Low"))  
  
JobRole Rank by Attrition =  
RANKX(ALL(Dim_JobRole[JobRole]), [Attrition Rate %], , DESC)
```

3. Outlier & Trend Analysis

Attrition rate was analyzed across every categorical dimension, and numeric fields were tested for outliers using the IQR (Interquartile Range) method.

Attrition Rate by Segment

Dimension	Highest-Risk Segment	Rate	Lowest-Risk Segment	Rate
OverTime	Yes	30.5%	No	10.4%
Job Role	Sales Representative	39.8%	Research Director	2.5%
Business Travel	Frequently	24.9%	Non-Travel	8.0%
Marital Status	Single	25.5%	Divorced	10.1%
Job Level	Level 1	26.3%	Level 4	4.7%
Stock Option Level	0	24.4%	2	7.6%
Work-Life Balance	1 (Poor)	31.2%	3 (Good)	14.2%
Department	Sales	20.6%	R&D	13.8%

Leavers vs. Stayers — Numeric Comparison

Metric	Stayers	Leavers	Difference
Monthly Income	\$6,833	\$4,787	-29.9%
Years in Current Role	4.5	2.9	-35.6%
Years With Current Manager	4.4	2.9	-34.1%
Total Working Years	11.9	8.2	-31.1%
Years at Company	7.4	5.1	-31.1%
Age	37.6	33.6	-10.6%
Distance From Home (mi)	8.9	10.6	+19.1%
Num. Companies Worked	2.6	2.9	+11.5%

IQR-Based Outliers

Field	Outlier Count	Attrition Rate Within Outliers
Monthly Income (very high earners)	114	4.4% — high pay strongly protects retention
Num. Companies Worked (6+ prior jobs)	52	23.1% — job-hoppers churn faster
Years Since Last Promotion (stalled 6-7+ yrs)	107	12.1% — moderate risk factor
Years at Company (18+ yrs, veterans)	104	9.6% — below-average risk

4. Narrative Insights & Recommendations

- Overtime is the single strongest driver — employees working overtime leave at nearly 3x the rate of those who don't (30.5% vs 10.4%). Action: audit workload distribution in Sales and Laboratory Technician roles.
- Entry-level compensation is a flight risk — Job Level 1 and zero stock options correlate strongly with attrition. Action: introduce retention bonuses or accelerated vesting in years 1–2.
- Sales Representatives are the highest-risk role at 39.8% attrition. Action: targeted retention program — career pathing, mentorship, and a compensation review.
- Early tenure (0–2 years) is the danger zone — most leavers exit before building tenure with a manager. Action: strengthen onboarding and early-career check-ins.
- Frequent travel and poor work-life balance compound risk — WLB=1 shows the highest attrition (31.2%). Action: review travel frequency and workload for at-risk roles.

5. SQL Query — Attrition Count by Department

Task: count employees who left the company (Attrition = 'Yes'), grouped by Department, sorted descending.

```
SELECT Department, COUNT(*) AS AttritionCount
FROM EmployeeAttrition
WHERE Attrition = 'Yes'
GROUP BY Department
ORDER BY AttritionCount DESC;
```

Department	AttritionCount
Research & Development	133
Sales	92
Human Resources	12

6. Power BI Dashboard — Page Guide

Page	Content
Home	Navigation cover page with links to all report pages.
Overview	KPI cards (Total Employees, Attrition Count, Attrition Rate %, OverTime Attrition Rate %), Attrition Rate % by Department, Attrition Count by OverTime, and an interactive Decomposition Tree.
Drivers	Attrition Rate % by JobRole, by MaritalStatus, by StockOptionLevel, and Avg Monthly Income by Department (Leavers vs Stayers).
Trends & Outliers	Decomposition Tree, JobSatisfaction × EnvironmentSatisfaction pivot, Income vs. Experience scatter plot by Attrition.
Recommendations	Narrative insights and actionable recommendations for stakeholders.